

Needle Cricothyrotomy

Needle cricothyrotomy is a temporizing intervention in adults and children <10 years old in whom a surgical airway (cricothyrotomy) would otherwise be indicated. It only provides oxygenation, not ventilation, and is only effective for a short period of time.

Indications for Needle Cricothyrotomy

- Failure of 2 attempts per credentialed provider at oral intubation in patient in need of emergent airway intervention (all attempts should have utilized alternative methods – different provider, bougie, video laryngoscope, etc.)

AND

- Unable to secure airway with alternate airway device

AND AT LEAST ONE OF THE FOLLOWING:

- Inability to ventilate
- Inability to oxygenate
- Inability to clear obstructed airway in timely manner
- Massive facial or oral trauma preventing BVM, intubation, or alternate airway placement as indicated by tactical situation

AND

- Unable to perform surgical cricothyrotomy OR surgical cricothyrotomy is contraindicated

Relative Contraindications

- Able to ventilate successfully with BVM or other alternative method
- Expected prolonged time to definitive care

Equipment needed for surgical cricothyrotomy

- 18 gauge angiocath for infants and young children
- 14 gauge angiocath for adolescents and adults
- 5 or 10 ml syringe
- 7.0 ETT BVM connector and 3 ml syringe with plunger removed OR 3.0 ETT BVM connector
- Suction
- Oxygen with tubing
- Method to secure tube

Needle Cricothyrotomy Procedure

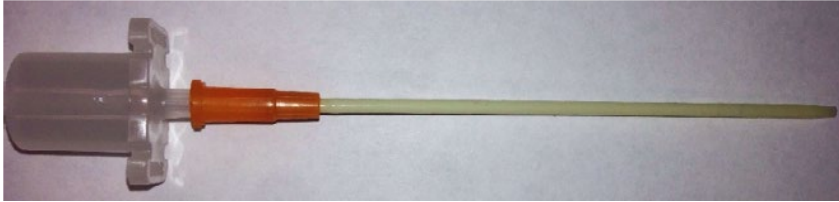
- 1) Prepare suction and cricothyrotomy equipment.
- 2) Place patient into neutral to sniffing position unless there is concern for trauma to c-spine.
- 3) Oxygenate the patient with 100% O₂ with BVM or NRB as appropriate.
- 4) Clean area with sterilizing prep, time permitting.
- 5) Identify the thyroid cartilage and stabilize it with your fingers.
- 6) Using a finger, identify cricothyroid membrane just below the thyroid cartilage, and above the cricoid cartilage.
- 7) Attach the 14- or 18 gauge IV catheter to a 5 or 10 ml syringe (1 ml of water in syringe aids in determining air return).
- 8) Puncture the skin in the midline over the cricothyroid membrane with the needle angled slightly toward patient's feet (a small incision with a scalpel may be made first to facilitate passage of the needle).
- 9) Maintain needle aspiration as the needle is inserted through the lower half of the cricothyroid membrane. Aspiration of air (or bubble in syringe if water used) indicates entrance into the trachea.
- 10) Change the angle of the syringe to 45°. Advance only the plastic catheter to the hub.
- 11) Remove the needle and syringe, leaving catheter in place. Maintain control of catheter until properly secured.
- 12) Attach the oxygen by one of the means noted below (see page II.15)
 - Place a 7.0 ETT connector on the end of a 3 cc syringe. Attach syringe to catheter.
 - Place a 3.0 ETT connector directly onto catheter.
- 13) Secure tube in place continue oxygenation with 100% O₂.
- 14) Attach BVM to adapter/syringe and gently ventilate.
***NOTE: a large amount of resistance will be felt due to the small catheter size
- 15) Confirm tube placement similar to orotracheal intubation.
- 16) Continuously monitor placement and functioning of tube.

Needle cricothyrotomy set-ups

Adults and adolescents – 14 gauge angiocath

Infants and children – 18 gauge angiocath

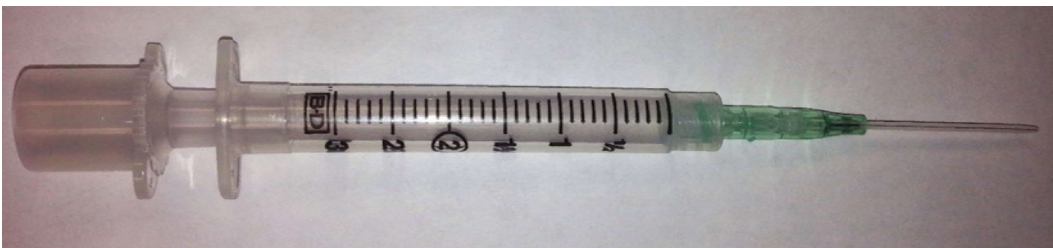
3.0 ETT BVM connector + 14 gauge angiocath



7.0 ETT BVM Connector + 3 ml syringe without stopper + 14 gauge angiocath



7.0 ETT BVM Connector + 3 ml syringe without stopper + 18 gauge angiocath



3.0 ETT BVM connector + 18 gauge angiocath

